

Beverage Quality Classification Using Random Forest and PCA: Technical Report

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1. Introduction

This study presents a classification analysis of beverage quality using supervised and unsupervised machine learning techniques. The data includes 4898 observations with 11 physicochemical numerical features and one categorical target variable: `quality`. The aim is to build a predictive model capable of distinguishing high-quality beverages (labelled "Excellent") from others.

2. Data Preparation and Preprocessing

- **Target Transformation:** The `quality` variable was converted to binary:
 - `Excellent` → 1
 - All other values → 0
 - **Feature Engineering:** Key variables (`alcohol`, `volatile acidity`, `sulphates`) were one-hot encoded to capture their categorical distribution.
 - **Scaling:** All features were standardized using `StandardScaler`.
 - **Class Imbalance:** The original class distribution was imbalanced (745 vs. 2683). SMOTE (Synthetic Minority Oversampling Technique) was applied to generate a balanced dataset with 2683 samples per class.
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3. Supervised Modeling

3.1 Decision Tree Classifier

- **Criterion:** Entropy
- **Max Depth:** 5
- **Accuracy:** 0.715
- **Confusion Matrix:**
 - TP: 215
 - TN: 836
 - FP: 319
 - FN: 100
- **Cross-validation Accuracy:** Mean = 0.724 ($\pm$ 0.022)

3.2 Random Forest Classifier

- **Initial Model:**
 - `n_estimators=300`, `max_features='sqrt'`, `criterion='entropy'`

- **Accuracy:** 0.883
 - **Grid Search Optimization:**
 - Best parameters: `n_estimators=450`, `max_depth=30`
 - Cross-validated score: 0.918
 - Final accuracy on test data: 0.881
 - **Feature Importance:** Top contributing features:
 - `density`: 0.401
 - `residual sugar`: 0.198
 - `citric acid`: 0.187
 - `free sulfur dioxide`: 0.093
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4. Unsupervised Learning: PCA and Clustering

4.1 Principal Component Analysis (PCA)

- Two components were extracted.
- **Explained Variance Ratio:** 1.89% total (Component 1: 1.09%, Component 2: 0.80%)
- PCA visualization showed clustering trends with several outliers and slight directional bias.

4.2 KMeans Clustering

- Elbow method identified **4** as the optimal number of clusters.
 - Clustering helped visualize how samples group based on principal components and validated feature selection.
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5. Recommendations

Based on the models and their interpretation:

- High-quality beverages tend to have:
 - **Alcohol content:** optimal at ~9.2–9.4
 - **Low volatile acidity, residual sugar, and sulphates**
 - The most predictive variable is **density**, followed by indicators of sugar and acidity.
 - Feature importance analysis suggests that after one-hot encoding, numerical distributions lose some predictive power in dummy formats; hence, transformations should be evaluated carefully.
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6. Conclusion

This project demonstrates the effective use of classification models and dimensionality reduction to predict product quality from chemical properties. Random Forest proved to be the most robust approach, particularly after parameter tuning and class balancing. PCA and KMeans provided valuable support for feature insight and unsupervised validation.

The methods used can be applied to other industrial datasets for quality control, especially where domain knowledge complements machine learning.

Keywords: Random Forest, SMOTE, Decision Tree, PCA, KMeans, beverage quality, classification, supervised learning, feature importance, model tuning.